

SCHOOL GARDEN DESIGN, INSTALLATION & TECHNICAL SPECIFICATION MANUAL

50 Level-Specific Learning Gardens | Nursery to SSS 1-3

Purpose	Provide preliminary school-scale design specifications for planning, budgeting, fabrication, installation and maintenance of the 50 GrowMeal learning gardens.
Design status	Educational prototype specifications. Site conditions, local codes, structural loads, water quality, accessibility requirements and electrical work must be verified before construction.
Scaling rule	Schools may scale footprints up or down while preserving safe access, learning function, treatment separation where relevant, and maintainable water supply.
Cost notation	Cost categories are indicative relative bands rather than market quotations; local procurement should establish actual budgets.

SYSTEM-WIDE DESIGN STANDARDS

Paths	Keep primary learner paths clear, stable and slip-resistant. Use wider access routes where required for mobility aids.
Bed reach	Avoid beds so wide that learners must step into them; typical school raised-bed width should remain about 1.0-1.2 m when accessed from both sides.
Water	Provide controllable water access, drainage and an isolation method. Avoid persistent standing water.
Structures	Anchor vertical frames, tables, tanks and supports against tipping. Cover sharp edges and protruding fasteners.
Materials	Prefer durable, repairable, locally available materials. Avoid treated/contaminated materials where food crops or frequent learner contact are expected.
Electrical/sensors	Use protected low-voltage learning equipment where possible. Qualified personnel should handle mains electrical installation.
Food-contact separation	Keep soil/garden tools separate from clean Ingredient Lab or Farm-to-Food utensils and containers.
Commissioning	Before learner use: inspect stability, water function, drainage, access, labels, tool control and emergency isolation.

GD-01 | Alphabet Garden

Nominal footprint	3 m x 4 m (adapt to site)
Nominal area	12 m2
Complexity	Low
Indicative cost band	Very low
Expected infrastructure life	1-2 seasons
Water approach	Hand watering / small cans
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; durable learning markers; clearly separated planting pockets
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-02 | Number Garden

Nominal footprint	3 m x 4 m (adapt to site)
Nominal area	12 m2
Complexity	Low
Indicative cost band	Very low
Expected infrastructure life	1-2 seasons
Water approach	Hand watering / small cans
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; durable learning markers; clearly separated planting pockets
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-03 | Geometry Garden

Nominal footprint	3 m x 4 m (adapt to site)
Nominal area	12 m2
Complexity	Low
Indicative cost band	Very low
Expected infrastructure life	1-2 seasons
Water approach	Hand watering / small cans
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; durable learning markers; clearly separated planting pockets
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required

DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-04 | Leaf Garden

Nominal footprint	3 m x 4 m (adapt to site)
Nominal area	12 m2
Complexity	Low
Indicative cost band	Very low
Expected infrastructure life	1-2 seasons
Water approach	Hand watering / small cans
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-05 | Butterfly Garden

Nominal footprint	3 m x 4 m (adapt to site)
Nominal area	12 m2
Complexity	Low
Indicative cost band	Very low
Expected infrastructure life	1-2 seasons
Water approach	Hand watering / small cans
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; flowering plant zone; observation boundary; no-spray learning zone
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-06 | ReBottle Garden

Nominal footprint	3 m x 4 m (adapt to site)
Nominal area	12 m2
Complexity	Low
Indicative cost band	Very low
Expected infrastructure life	1-2 seasons
Water approach	Hand watering / small cans

Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; clean reused food-safe bottles; secure support/rail; drainage holes
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-07 | MicroGrow Garden

Nominal footprint	3 m x 4 m (adapt to site)
Nominal area	12 m2
Complexity	Low
Indicative cost band	Very low
Expected infrastructure life	1-2 seasons
Water approach	Hand watering / small cans
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; stable child-height table/bench; small trays/containers
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-08 | Mini Table Garden

Nominal footprint	3 m x 4 m (adapt to site)
Nominal area	12 m2
Complexity	Low
Indicative cost band	Very low
Expected infrastructure life	1-2 seasons
Water approach	Hand watering / small cans
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; stable child-height table/bench; small trays/containers
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-09 | Sack Garden

Nominal footprint	3 m x 4 m (adapt to site)
Nominal area	12 m2

Complexity	Low
Indicative cost band	Very low
Expected infrastructure life	1-2 seasons
Water approach	Hand watering / small cans
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; stable grow sacks/bags; spacing markers; drainage base
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-10 | Discovery Garden

Nominal footprint	3 m x 4 m (adapt to site)
Nominal area	12 m ²
Complexity	Low
Indicative cost band	Very low
Expected infrastructure life	1-2 seasons
Water approach	Hand watering / small cans
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-11 | Alphabet Food Garden

Nominal footprint	4 m x 6 m (adapt to site)
Nominal area	24 m ²
Complexity	Low-Moderate
Indicative cost band	Low
Expected infrastructure life	2-4 seasons
Water approach	Watering cans or simple hose/drip
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; durable learning markers; clearly separated planting pockets
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-12 | Number Food Garden

Nominal footprint	4 m x 6 m (adapt to site)
Nominal area	24 m ²
Complexity	Low-Moderate
Indicative cost band	Low
Expected infrastructure life	2-4 seasons
Water approach	Watering cans or simple hose/drip
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; durable learning markers; clearly separated planting pockets
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-13 | Geometry Garden

Nominal footprint	4 m x 6 m (adapt to site)
Nominal area	24 m ²
Complexity	Low-Moderate
Indicative cost band	Low
Expected infrastructure life	2-4 seasons
Water approach	Watering cans or simple hose/drip
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; durable learning markers; clearly separated planting pockets
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required

DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-14 | Leaf Garden

Nominal footprint	4 m x 6 m (adapt to site)
Nominal area	24 m2
Complexity	Low-Moderate
Indicative cost band	Low
Expected infrastructure life	2-4 seasons
Water approach	Watering cans or simple hose/drip
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-15 | Pollinator Garden

Nominal footprint	4 m x 6 m (adapt to site)
Nominal area	24 m2
Complexity	Low-Moderate
Indicative cost band	Low
Expected infrastructure life	2-4 seasons
Water approach	Watering cans or simple hose/drip
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; flowering plant zone; observation boundary; no-spray learning zone
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-16 | Raised Food Garden

Nominal footprint	4 m x 6 m (adapt to site)
Nominal area	24 m2
Complexity	Low-Moderate
Indicative cost band	Low
Expected infrastructure life	2-4 seasons
Water approach	Watering cans or simple hose/drip

Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; 2-4 raised beds, approx. 1.0-1.2 m wide; bed edging 250-450 mm high
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-17 | ReBottle Wall Garden

Nominal footprint	4 m x 6 m (adapt to site)
Nominal area	24 m2
Complexity	Low-Moderate
Indicative cost band	Low
Expected infrastructure life	2-4 seasons
Water approach	Watering cans or simple hose/drip
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; stable vertical support frame; secured planting containers; drainage capture/control; clean reused food-safe bottles; secure support/rail; drainage holes
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-18 | Sack GrowBag Garden

Nominal footprint	4 m x 6 m (adapt to site)
Nominal area	24 m2
Complexity	Low-Moderate
Indicative cost band	Low
Expected infrastructure life	2-4 seasons
Water approach	Watering cans or simple hose/drip
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; stable grow sacks/bags; spacing markers; drainage base
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-19 | Ground Production Garden

Nominal footprint	4 m x 6 m (adapt to site)
Nominal area	24 m2

Complexity	Low-Moderate
Indicative cost band	Low
Expected infrastructure life	2-4 seasons
Water approach	Watering cans or simple hose/drip
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; production rows/beds; harvest access; crop labels
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-20 | Discovery Production Garden

Nominal footprint	4 m x 6 m (adapt to site)
Nominal area	24 m2
Complexity	Low-Moderate
Indicative cost band	Low
Expected infrastructure life	2-4 seasons
Water approach	Watering cans or simple hose/drip
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-21 | Raised Food Garden

Nominal footprint	5 m x 8 m (adapt to site)
Nominal area	40 m ²
Complexity	Moderate
Indicative cost band	Moderate
Expected infrastructure life	3-5 years with maintenance
Water approach	Gravity drip / controlled watering
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; 2-4 raised beds, approx. 1.0-1.2 m wide; bed edging 250-450 mm high
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-22 | Geometry Garden

Nominal footprint	5 m x 8 m (adapt to site)
Nominal area	40 m ²
Complexity	Moderate
Indicative cost band	Moderate
Expected infrastructure life	3-5 years with maintenance
Water approach	Gravity drip / controlled watering
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; durable learning markers; clearly separated planting pockets
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-23 | Grid Experiment Garden

Nominal footprint	5 m x 8 m (adapt to site)
Nominal area	40 m ²
Complexity	Moderate
Indicative cost band	Moderate
Expected infrastructure life	3-5 years with maintenance
Water approach	Gravity drip / controlled watering
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; marked experimental plots; treatment labels; buffer strips; measurement points
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required

DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-24 | Keyhole Garden

Nominal footprint	5 m x 8 m (adapt to site)
Nominal area	40 m ²
Complexity	Moderate
Indicative cost band	Moderate
Expected infrastructure life	3-5 years with maintenance
Water approach	Gravity drip / controlled watering
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; circular/near-circular raised bed; central compost basket; keyhole access path
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-25 | Compost Garden

Nominal footprint	5 m x 8 m (adapt to site)
Nominal area	40 m ²
Complexity	Moderate
Indicative cost band	Moderate
Expected infrastructure life	3-5 years with maintenance
Water approach	Gravity drip / controlled watering
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; 2-3 compost bays/bins; carbon/nitrogen material station; temperature/moisture observation point
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-26 | Gravity Drip Garden

Nominal footprint	5 m x 8 m (adapt to site)
Nominal area	40 m ²
Complexity	Moderate
Indicative cost band	Moderate
Expected infrastructure life	3-5 years with maintenance

Water approach	Gravity drip / controlled watering
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; header tank where appropriate; mainline/laterals; valves; drip emitters or measured outlets
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-27 | Vertical Garden

Nominal footprint	5 m x 8 m (adapt to site)
Nominal area	40 m2
Complexity	Moderate
Indicative cost band	Moderate
Expected infrastructure life	3-5 years with maintenance
Water approach	Gravity drip / controlled watering
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; stable vertical support frame; secured planting containers; drainage capture/control
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-28 | Crop Rotation Garden

Nominal footprint	5 m x 8 m (adapt to site)
Nominal area	40 m2
Complexity	Moderate
Indicative cost band	Moderate
Expected infrastructure life	3-5 years with maintenance
Water approach	Gravity drip / controlled watering
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-29 | Water-Smart Garden

Nominal footprint	5 m x 8 m (adapt to site)
Nominal area	40 m2

Complexity	Moderate
Indicative cost band	Moderate
Expected infrastructure life	3-5 years with maintenance
Water approach	Gravity drip / controlled watering
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; header tank where appropriate; mainline/laterals; valves; drip emitters or measured outlets; sensor mounting points; protected low-voltage enclosure; manual override; data-record station
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-30 | Intro Soilless Garden

Nominal footprint	5 m x 8 m (adapt to site)
Nominal area	40 m2
Complexity	Moderate
Indicative cost band	Moderate
Expected infrastructure life	3-5 years with maintenance
Water approach	Gravity drip / controlled watering
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; food-grade reservoir; channels/containers; growing medium/net pots; return/drainage path
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-31 | Production Garden

Nominal footprint	6 m x 10 m (adapt to site)
Nominal area	60 m ²
Complexity	Moderate-Advanced
Indicative cost band	Moderate-High
Expected infrastructure life	3-7 years
Water approach	Measured irrigation / tanks / optional sensors
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; production rows/beds; harvest access; crop labels
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-32 | Raised Systems Garden

Nominal footprint	6 m x 10 m (adapt to site)
Nominal area	60 m ²
Complexity	Moderate-Advanced
Indicative cost band	Moderate-High
Expected infrastructure life	3-7 years
Water approach	Measured irrigation / tanks / optional sensors
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; 2-4 raised beds, approx. 1.0-1.2 m wide; bed edging 250-450 mm high; modular test stations; measurement bench; system map/signage
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-33 | Grid Experiment Garden

Nominal footprint	6 m x 10 m (adapt to site)
Nominal area	60 m ²
Complexity	Moderate-Advanced
Indicative cost band	Moderate-High
Expected infrastructure life	3-7 years
Water approach	Measured irrigation / tanks / optional sensors
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; marked experimental plots; treatment labels; buffer strips; measurement points
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required

DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-34 | Crop Rotation Garden

Nominal footprint	6 m x 10 m (adapt to site)
Nominal area	60 m2
Complexity	Moderate-Advanced
Indicative cost band	Moderate-High
Expected infrastructure life	3-7 years
Water approach	Measured irrigation / tanks / optional sensors
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-35 | Intercropping Garden

Nominal footprint	6 m x 10 m (adapt to site)
Nominal area	60 m2
Complexity	Moderate-Advanced
Indicative cost band	Moderate-High
Expected infrastructure life	3-7 years
Water approach	Measured irrigation / tanks / optional sensors
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-36 | Water-Smart Garden

Nominal footprint	6 m x 10 m (adapt to site)
Nominal area	60 m2
Complexity	Moderate-Advanced
Indicative cost band	Moderate-High
Expected infrastructure life	3-7 years
Water approach	Measured irrigation / tanks / optional sensors

Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; header tank where appropriate; mainline/laterals; valves; drip emitters or measured outlets; sensor mounting points; protected low-voltage enclosure; manual override; data-record station
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-37 | Compost & Soil Garden

Nominal footprint	6 m x 10 m (adapt to site)
Nominal area	60 m2
Complexity	Moderate-Advanced
Indicative cost band	Moderate-High
Expected infrastructure life	3-7 years
Water approach	Measured irrigation / tanks / optional sensors
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; 2-3 compost bays/bins; carbon/nitrogen material station; temperature/moisture observation point
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-38 | Soilless Learning Garden

Nominal footprint	6 m x 10 m (adapt to site)
Nominal area	60 m2
Complexity	Moderate-Advanced
Indicative cost band	Moderate-High
Expected infrastructure life	3-7 years
Water approach	Measured irrigation / tanks / optional sensors
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; food-grade reservoir; channels/containers; growing medium/net pots; return/drainage path
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-39 | Sensor-Ready Garden

Nominal footprint	6 m x 10 m (adapt to site)
Nominal area	60 m2

Complexity	Moderate-Advanced
Indicative cost band	Moderate-High
Expected infrastructure life	3-7 years
Water approach	Measured irrigation / tanks / optional sensors
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; sensor mounting points; protected low-voltage enclosure; manual override; data-record station
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-40 | Applied Systems Garden

Nominal footprint	6 m x 10 m (adapt to site)
Nominal area	60 m ²
Complexity	Moderate-Advanced
Indicative cost band	Moderate-High
Expected infrastructure life	3-7 years
Water approach	Measured irrigation / tanks / optional sensors
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; modular test stations; measurement bench; system map/signage
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-41 | Advanced Production Garden

Nominal footprint	8 m x 12 m (adapt to site)
Nominal area	96 m ²
Complexity	Advanced school demonstrator
Indicative cost band	High
Expected infrastructure life	5-10 years for durable infrastructure
Water approach	Instrumented irrigation / tanks / control options
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; production rows/beds; harvest access; crop labels
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-42 | Crop Trial Garden

Nominal footprint	8 m x 12 m (adapt to site)
Nominal area	96 m ²
Complexity	Advanced school demonstrator
Indicative cost band	High
Expected infrastructure life	5-10 years for durable infrastructure
Water approach	Instrumented irrigation / tanks / control options
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; marked experimental plots; treatment labels; buffer strips; measurement points
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-43 | Controlled Experiment Garden

Nominal footprint	8 m x 12 m (adapt to site)
Nominal area	96 m ²
Complexity	Advanced school demonstrator
Indicative cost band	High
Expected infrastructure life	5-10 years for durable infrastructure
Water approach	Instrumented irrigation / tanks / control options
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; marked experimental plots; treatment labels; buffer strips; measurement points
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required

DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-44 | Irrigation Engineering Garden

Nominal footprint	8 m x 12 m (adapt to site)
Nominal area	96 m2
Complexity	Advanced school demonstrator
Indicative cost band	High
Expected infrastructure life	5-10 years for durable infrastructure
Water approach	Instrumented irrigation / tanks / control options
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; header tank where appropriate; mainline/laterals; valves; drip emitters or measured outlets
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-45 | Water Productivity Garden

Nominal footprint	8 m x 12 m (adapt to site)
Nominal area	96 m2
Complexity	Advanced school demonstrator
Indicative cost band	High
Expected infrastructure life	5-10 years for durable infrastructure
Water approach	Instrumented irrigation / tanks / control options
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-46 | Advanced Soilless Garden

Nominal footprint	8 m x 12 m (adapt to site)
Nominal area	96 m2
Complexity	Advanced school demonstrator
Indicative cost band	High
Expected infrastructure life	5-10 years for durable infrastructure
Water approach	Instrumented irrigation / tanks / control options

Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; food-grade reservoir; channels/containers; growing medium/net pots; return/drainage path
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-47 | Sensor & Monitoring Garden

Nominal footprint	8 m x 12 m (adapt to site)
Nominal area	96 m2
Complexity	Advanced school demonstrator
Indicative cost band	High
Expected infrastructure life	5-10 years for durable infrastructure
Water approach	Instrumented irrigation / tanks / control options
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; sensor mounting points; protected low-voltage enclosure; manual override; data-record station
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-48 | Smart Irrigation Demonstrator

Nominal footprint	8 m x 12 m (adapt to site)
Nominal area	96 m2
Complexity	Advanced school demonstrator
Indicative cost band	High
Expected infrastructure life	5-10 years for durable infrastructure
Water approach	Instrumented irrigation / tanks / control options
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; header tank where appropriate; mainline/laterals; valves; drip emitters or measured outlets; sensor mounting points; protected low-voltage enclosure; manual override; data-record station
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-49 | Research & Design Garden

Nominal footprint	8 m x 12 m (adapt to site)
Nominal area	96 m2

Complexity	Advanced school demonstrator
Indicative cost band	High
Expected infrastructure life	5-10 years for durable infrastructure
Water approach	Instrumented irrigation / tanks / control options
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; modular test stations; measurement bench; system map/signage
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

GD-50 | Integrated Food Systems Laboratory

Nominal footprint	8 m x 12 m (adapt to site)
Nominal area	96 m2
Complexity	Advanced school demonstrator
Indicative cost band	High
Expected infrastructure life	5-10 years for durable infrastructure
Water approach	Instrumented irrigation / tanks / control options
Core components	level-specific learning sign; defined learner pathway; planting/growing zone; water access; tool/material station; modular test stations; measurement bench; system map/signage
Preliminary BOM	timber/blocks/metal edging as appropriate; clean growing medium; compost; planting material; labels; watering/irrigation fittings; hand tools; mulch; fasteners; PPE as required
DIY/local option	Use locally available untreated durable timber, masonry blocks, food-safe reused containers, buckets/tanks and simple gravity irrigation where safe and maintainable.
Installation sequence	Set out and level -> establish paths/boundaries -> build beds/supports -> install water/drainage -> add growing medium -> label learning zones -> test safety/function -> plant -> commission for teaching.
Safety/commissioning	Rounded/covered edges; stable structures; clear paths; no protruding fasteners; controlled tools; drainage; supervised water points; age-appropriate working heights.
Maintenance	Weekly condition check; irrigation/water check; path/structure inspection; seasonal growing-medium/crop reset; repair defects before learner access.

APPENDIX A - PRE-INSTALLATION SITE SURVEY

School/site	_____
Proposed level/garden	_____
Available footprint	_____
Water source/distance	_____
Drainage condition	_____
Ground/surface condition	_____
Sun/shade observation	_____
Access/accessibility	_____
Existing hazards/services	_____
Storage/tool control	_____
Required adaptations	_____
Approved by/date	_____

APPENDIX B - INSTALLATION QUALITY CHECK

Check	Pass	Action / Notes
Dimensions/set-out verified		
Paths clear and stable		
Beds/containers stable		
Vertical structures anchored		
Tank/water point stable		
Irrigation tested		
Drainage works		
No sharp/protruding hazards		
Growing medium acceptable		
Labels installed		
Tool storage ready		
Learning station usable		
Accessibility adaptations complete		
Electrical/sensor enclosure protected where applicable		
Final safety inspection complete		

APPENDIX C - PRELIMINARY COST PLAN

Site preparation	_____
Beds/structures	_____
Water/irrigation	_____
Growing media/compost	_____
Seeds/planting materials	_____
Tools/storage	_____
Measurement/sensors	_____
Learning signage	_____
Labor/fabrication	_____
Contingency	_____
Total	_____